

3. Optimal enrolment

What is $dp_table[i][j]$:

- Expected no. units passed using the first i units, with at most j hours
- Unit i in best selection, or it is not.

What is 1.30?

- The expected no. passed units
- It is an average, not a guarantee

Time + space complexities?

- Time: $O(\text{units} \times \text{hours})$
 - Not polynomial: is pseudo-polynomial
- Space: $O(\text{units} \times \text{hours})$

4. Similarity score

What is $dp_table[i][j]$; why is row 0, column 0 = 0?

- $Dp_table[i][j]$ is the length of the LCS
- We measure the LCS between two strings A and B , of size i , and j
- $Row = Column = 0$ represents the base case; passing an empty string must return an LCS of size 0; it also drives the logic for taking the of the top left diagonal cell upon a match.

Greedy approach: scan A , for each character, match to the earliest usable occurrence in B .

- $A = xab, B = abx$:
 - x matches to last slot; returns 1
 - Nothing remains for a, b

Why is it safe to take left, or upper, but not diagonal?

- Up and left are always at least equal to diagonal
- Non-decreasing; $\therefore \max(up, left) \geq diagonal$
- When two chars distinct, at least one is not in the LCS.

5. One-way travel

In 0-1 BFS, why do free roads go to the front of the queue, and demerit roads go to the back?

- deque stays sorted by distance; holds distance d at front, and $d + 1$ at the back
 - Free road: road to node at distance d , hence front
 - Demerit road: leads to a node at distance $d + 1$, hence back
 - Nodes appear in non-decreasing order

Outbound and return trip is separately computed; what assumption about how demerits work makes that valid?

- Two legs are independent
 - Wrong trips cost a demerit *each time*
 - The initial trip does not affect the second
- Rule break: “once driven the wrong way, the trip becomes free onwards”

Time complexity? Can a node be popped more than once in 0-1 BFS? Does it matter?

- Time: $O(V + E)$
 - Note: parsing is non-quadratic:
 - Inner loop: runs once per number in the input
- Node can be popped more than once; this happens if the distance is improved after the enqueue
 - 0-1 BFS occurs at most twice per node